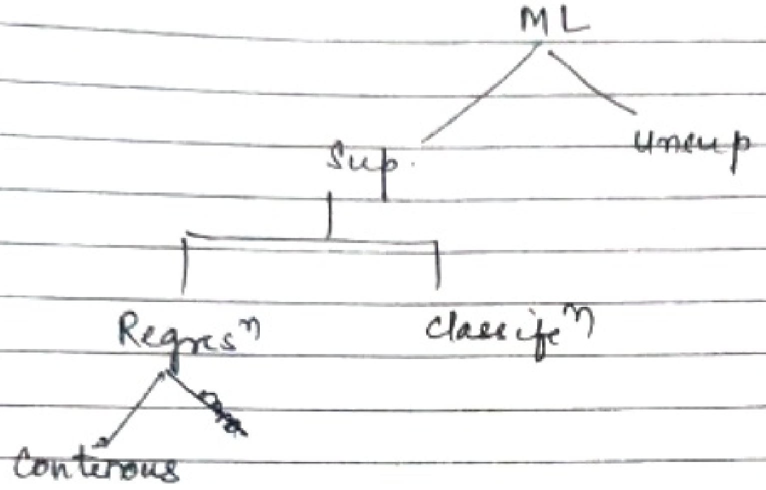
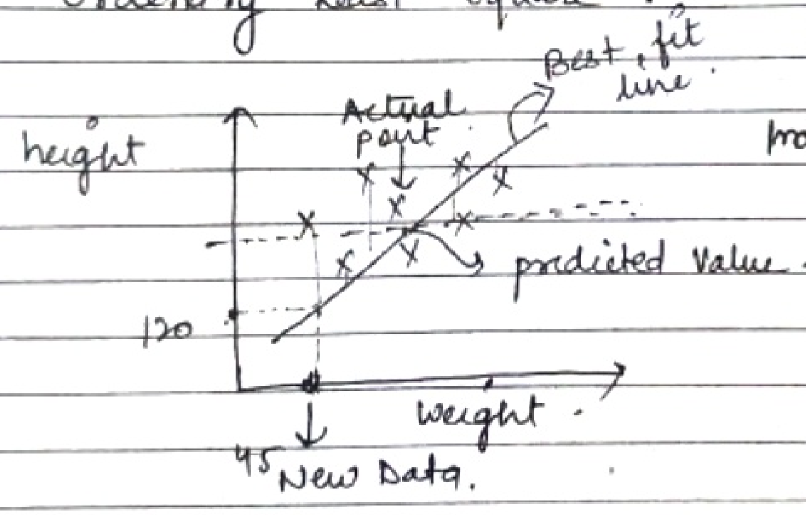
Linear Regression

Problem we are solving



We can solve Regression by statistically called as Ordinary Least Square.



Trained Data

Indep (kg) weight	Dep. (cm) height
73	150
63	155
78	170
82	165

can be Contin.

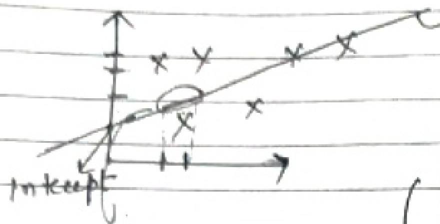
Train
↓
Model
↓
I/P weight → $\boxed{\text{hypothesis}}$
↓
height
↓
O/P

Residual Error :- Distance b/w Actual points from Best fit line.
Diff b/w Actual & predicted Value.

Aim :- Submission of all those pts - Distance should be minimal.
than only said Best fit.

Aim :- To find Best Fit Line with Minimal Error

Solⁿ :- ① (Basic maths)



$\hat{y} = mx + c$

$\hat{y} \rightarrow$ predicted pt

$m \rightarrow$ slope or coeff

Infomⁿ — slope kitna ↑

(1 unit Increase in $x \uparrow \rightarrow$ ↑ y unit y kitna increase hora hai)

c = Intercept

means if $x = 0$, then in which pt y get it value.

(Jab $x=0$, konse pt me y ko milra hai.)

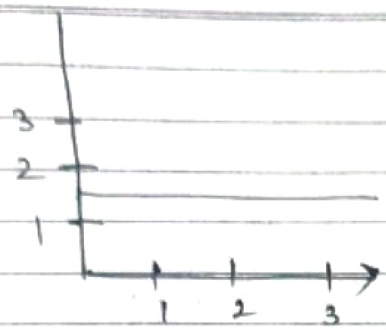
In other words :- If $x=0$ then where is the Best Fit line Intercepting zero, is c.

$h_\theta(x)^{(i)} = \theta_0 + \theta_1 x_1^{(i)}$ $\quad x \rightarrow$ data points.

$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \cdots \theta_n x_n$.

hypothesis $\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$

(y is a linear functⁿ of x)



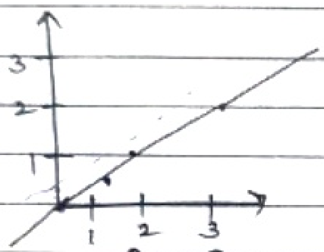
$h(\theta)x^i = \theta_0 + \theta_1 x_1$

$\Rightarrow 1.5 + 0$

$\Rightarrow$

$\theta_0 = 1.5$

$\theta_1 = 0.$



$\theta_0 = 0$

$\theta_1 = 1.5$

Intercept, coeff

$h_\theta(x) = \theta_0 + \theta_1 x_1$

$h_\theta(1) \Rightarrow \theta_0 + 1.5(1) = 0.5$

$h_\theta(2) = 0 + 1.5(2) = 1.0$

$h_\theta(3) = 0 + 1.5(3) = 3.$

Cost funct^n (Minimise θ_0, θ_1)

$\sum_{i=1}^{m} \frac{1}{2m} \left(h_\theta(x)^{(i)} - y^{(i)} \right)^2$

predicted pt — Actual pt.

(why square) if $h_\theta(x)$ is min than it comes -ve value.

minimise θ_0, θ_1

$$\boxed{\Rightarrow \sum_{i=1}^{m} \frac{1}{2m} \left(h_\theta(x)^i - y^{(i)} \right)^2}$$

$m \rightarrow$ no. of data points.

why $\frac{1}{m}$? to find out Average of all all errors.

why 2? x^2

why Differentiat^n $\Rightarrow \frac{\partial(x^2)}{\partial x} = 2x^{2-1}$

$\Rightarrow 2x$

Sol^n to find out slope.

Naive Bayes

No correlat^n

Delta

Cost funct^n = $J(\theta_0,\theta_1)$ ↓↓

$$\boxed{J(\theta_0,\theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_{\theta_0}(x)^{i} - y^{(i)}\right)^2}$$

↳ Squared Error funct^n

hypo	Cost funct
$h_\theta(x) = \theta_0 + \theta_1 x_1$	$J(\theta_0,\theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x_i) - y_i\right)^2$

Let $\theta_0 = 0$.

$h_\theta(x) = \theta_1 x_1$.

how we will change θ_1 value.

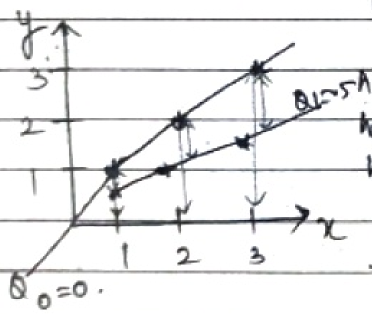
optimizer

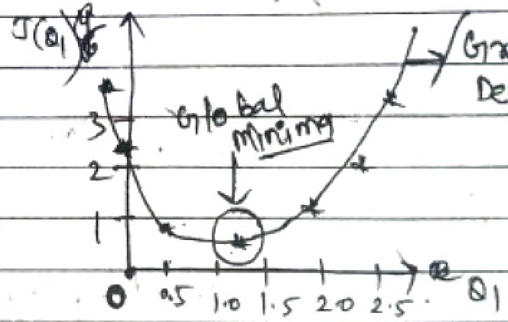
when $\theta_1 = 1$ (1,1) (2,2) (3,3)

$\theta_1 = .5$ $h_\theta(x) = 1\times1$

$h_\theta(x) = 1\times2$

$h_\theta(x) = 1\times3$





$J(\theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x)^i - y^i\right)^2$

$\theta_1 = 1$ 2×3

$\Rightarrow \frac{1}{6}\left(0^2 + 0^2 + 0^2\right)$

$\Rightarrow \underline{0}$.

$\theta_1 = 0.5$

$\theta_1 x_1 = 0.5\times1 = .5$

$\Rightarrow .5\times2 = 1.0$

$= .5\times3 = 1.5$

$\theta_1 = 0.5 \Rightarrow J(\theta_1) = \frac{1}{6}\left[(0.5-1)^2 + (1-2)^2 + (1.5-3)^2\right]$

$\Rightarrow \frac{1}{6}(3.5) \approx \underline{0.58}$

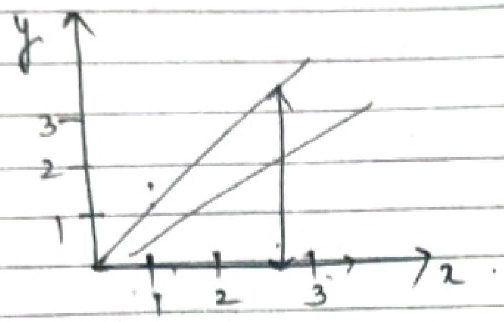
BFL = best fit line

1 BFL - 1 BFL

Naik — no correlat

Delta

$\theta_1 = 0.0$

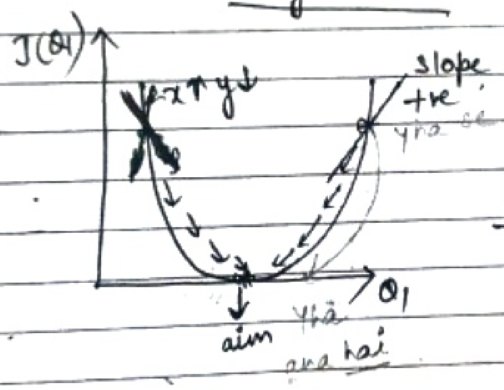


$$J(\theta_1) = \frac{1}{6}\left[(0-1)^2 + (0-2)^2 + (0-3)^2\right]$$

$$\Rightarrow \frac{14}{6} \approx \underline{2.3}$$

Global Minima → point till you change the θ_1 value or train your point
(point ~~that~~ where error is very minimal ≈ 0)

<u>why GD?</u> Kaise pta chlega ki Global Minima aagya? Kitni bar Intialize?



-ve x↑ y↓

<u>For that we use</u>

Repeat Convergence Algo / Theorm

θ_1 → update → find derivate or slope

$$\left\{ \theta_j = \theta_j - \alpha \boxed{\frac{\partial}{\partial \theta_j}(J\theta_1)} \right.$$

learning rate — α ; slope

$$\boxed{\theta_1 = \theta_1 - \alpha(+ve)}$$ alway θ_1 decrease

$$\theta_1 = \theta_1 - \alpha(-ve) \; \}$$

Increment hoga. $\boxed{\theta_1 \Rightarrow \theta_1 +}$

Naive Bayes

No correlat^n

Learning rate = small (slow movement)
⇒ Large (jump movement)
(It might possible that it will not reach global minima)

<u>Outline</u>

① Start with θ_0 & θ_1

② Keep changing $\theta_0, \theta_1 \rightarrow$ reduce $(J(\theta_0, \theta_1))$ until we reach near global minima.

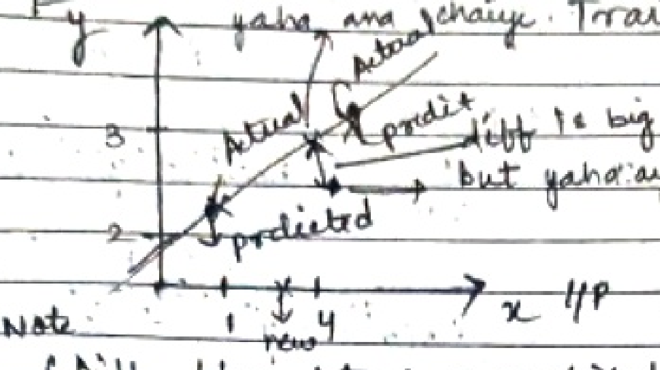
③ Convergence theorem.

$$\left\{ \theta_j := \theta_j - \alpha \frac{\partial}{\partial_j}\left(J(\theta_0, \theta_1)\right) \right\} \quad j = 0, \text{and } 1.$$

Worke on Overfitting problem of linear Regression

Ridge and lasso Regression

based on new pt o/p
yaha ana chahiye. Training set



x	y
1	2
4	3

Linear Reg

Note
(Diff b/w Actual & predicted ≈ 0)

New Data point comes

Overfitting

Train Data	~~Test Data~~
Train Accuracy = 90% ↑	Low Bias
Test Accuracy = 70% ↓↓ Predicted value o/p decreases	↓ - high Variance

$\boxed{\text{This is known as Overfitting}}$

Underfitting

Train Accuracy :- 60%
Test Accuracy = 62%
} Accuracy is bad

high bias & high Variance.

* Target.

Generalised Model :- Low Variance & Low Bias.

Train - 90
Test - 89%.

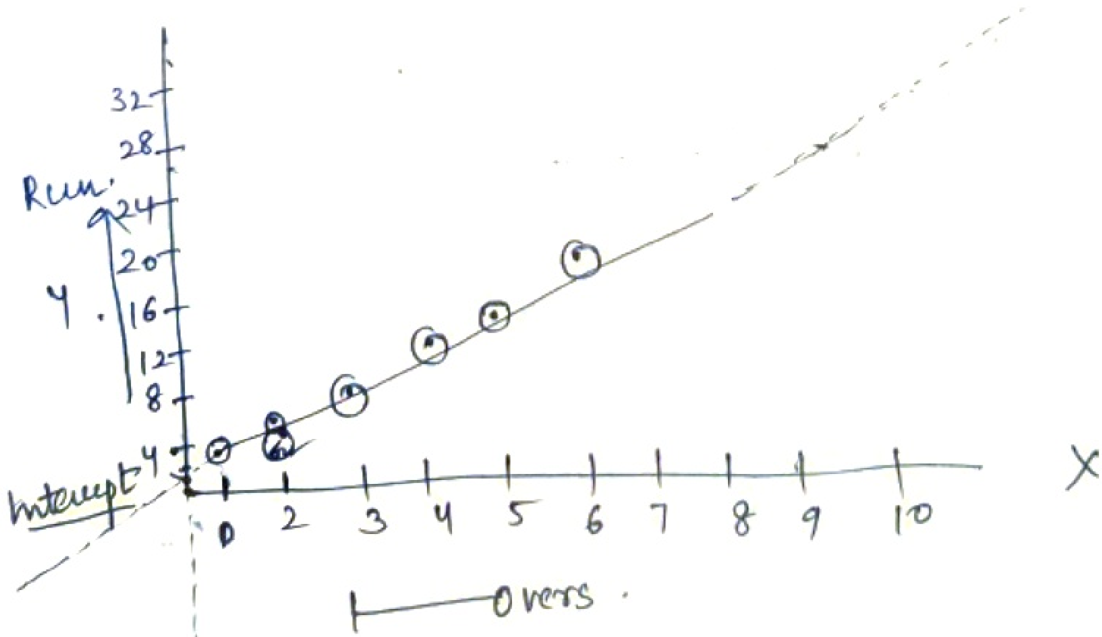
Regression Analysis

1. Co-relation analysis established the fact that many times two variables are closely related

2. Now we may be interested in estimating (predicting) the value of one variable given value of Another.

Overs (x)	Runs (y)
1	2
2	5
3	8
4	12
5	14
6	18
7	
8	
9	

Regression Analysis Reveals Average Relationship b/w two variable and this makes possible estimation/prediction.



Overs (X)

→ Independent variable / Explanatory variable

→ Used to predict the variable of interest (Y)

Runs (Y)

→ Dependent variables / Explained variable.

→ It is predicted by using explanotory variable.

Analysis used is called.

Simple Linear Regression. Analysis

Regression Analysis

- Linear Regression
- Multiple Linear Regression
- Non-Linear Regression.

The simple linear eqn expressed using following eqn :-

$$Y = a + bX + \varepsilon.$$

Y = dep. variable

X = Indep. variable

a = Intercept

b = slope

ε = Residual error.

Why method of least squares?

$\Sigma(Y - Y_c)^2 \rightarrow$ It should be minimum.

(Y → Actual, Y_c → Calculated)

Best fit

$$\boxed{\Sigma(Y - Y_c)^2 = 0}$$

For determining the value of 'a' & 'b' simultaneously

$\Sigma Y = Na + b\Sigma X$ $N \rightarrow$ no. of obsrⁿ

$\Sigma XY = a\Sigma X + b\Sigma X^2$

Regression eqⁿ of X on Y

$$X = a + bY$$

For determining the value of 'a' & 'b'

$\Sigma X = Na + b\Sigma Y$

$\Sigma XY = a\Sigma Y + b\Sigma Y^2$

Find

Y on X

$$Y = a + bX.$$

X	Y	XY	X^2	Y^2
1	2	2	1	4
2	5	10	4	25
3	8	24	9	64
4	12	48	16	144
5	14	70	25	196
6	18	108	36	324
Σ21	Σ59.	Σ262	Σ91	Σ757

$\Sigma Y = Na + b\Sigma X$

$\Sigma XY = a\Sigma X + b\Sigma X^2$

put values

$(59 = 6a + b21) \times 7$ — (1)

$(262 = a \times 21 + b \times 91) \times 2$ — (2)

$\Rightarrow 413 = 42a + 147b$ — (3)

$\Rightarrow 524 = 42a + 182b$ — (4)

$-111 = -35b$

$b = \frac{-111}{-35} = 3.17.$

Put Value of ~~b~~ in eq (1)

$\Rightarrow \cancel{b}\ a = -1.26.$

$$\boxed{Y = -1.26 + 3.17X}$$

Now let assume

$X = 12$ over.

$Y = -1.26 + 3.17 \times 12$

$\Rightarrow -1.26 + 38.04$

$\Rightarrow$ 36.78 Runs.

ANCOVA

(Analysis of covariance)

ANOVA + COVARIATE = ANCOVA

Analysis of Covariance (ANCOVA) provides a way of statiscally controlling the (linear) effect of variables one does not want to examine in a study.

→ These extraneous / confounding variables are called covariates.

Extraneous Variable

An extraneous variable is any variable not being Investigated that has the potential to affect the outcome of a research study.

In other words, it is any factor not considered as Independent variable that can affect the dep. variable or controlled conditions.

example :- Pre Test scores are used as Covariate in pre-& post test design.

ANOVA → Group mean compare.

ANCOVA → Adjusted Mean compare (covariate)

ANCOVA is used in examining the differences in the mean values of the dependent variable that are related to the effect of the controlled independent variable while taking into the account the influence of the uncontrolled indep variable.

	Teaching Method — Indep.		
	Lecture Method.	Lecture-cum Demonstration	Video based teaching
Intelligence (uncontrolled Ind. var)	A	B	C
Achievement (Dep)	X	Y	Z

$$\underbrace{X_n \qquad Y_n \qquad Z_n}_{\text{Adjusted Values}}$$

then ANCOVA compare these adjusted values and conclude.

ASSUMPTIONS for one-way ANCOVA

1. Dependent variable and covariate variable(s) should be measured on a continuous scale (interval or ratio level).

2. Indep. variable should consist of two or more categorical, indep. groups.

3. Observation should be independent.

4. Approximately Normally Distributed. Data for each category of indep. variable. (Shapiro-Wilk Test)

(5) Homogenity of Variances (Levene's Test).

(6) Covariate should be literally related to the dependent variable at each level of the Independent variable.
(group scatterplot → covariate).

(7) There should be no significant outliers.

(8) Homoscedasticity (having the same scatter)

d

GAUSS MARKOV Theorem

Regression Model

→ Given a vector of inputs $X_{n\times p} = ((x_{ij}))$, we predict output y via model.

$$\boxed{Y_{n\times 1} = X_{n\times p}\, \beta_{p\times 1} + \epsilon_{n\times 1}}$$

n = no of sample
p = no. of predictors
y = response vector.

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n\times 1}, \quad X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & & & \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}_{n\times p}, \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}_{n\times 1}.$$

→ $X_{n\times p}$ known as design matrix typically are considered as deterministic and $n > p$.

Because $(X^T X)_{p\times p}$
↳ full Rank matrix

and we got unique solution.

→ ϵ, (Epsilon) (also known as error/residuals) for all i are random variables, $i = 1, 2, \ldots n$.

1. $E(\epsilon_i) = 0, \ \forall i$ → Expectation
2. $Var(\epsilon_i) = E(\epsilon_i^2) = \sigma^2, \ \forall i$ Homoscedasticity — for all sample
3. $Cov(\epsilon_i, \epsilon_j) = E(\epsilon_j \epsilon_i) = 0, \ \forall i \neq j$ Independence.

This we can write in matrix Notation.

1. $E(\epsilon) = O_n$ — All element are zero.
2. $Cov(\epsilon) = \sigma^2 I_n$ → Diagonal matrix $\begin{bmatrix} \sigma^2 & 0 & 0 \\ 0 & \sigma^2 & 0 \\ 0 & 0 & \sigma^2 \end{bmatrix}$

estimator $\hat{\beta}$ is an unbiased of β.

Implication of the Assumption

→ Assumption

1. $E(\epsilon) = 0_n$
2. $Cov(\epsilon) = \sigma^2 I_n$.

→ It Induces distribution on y such that

$$E(y) = E(X\beta + \epsilon) = X\beta + \overset{0}{E(\epsilon)} = X\beta.$$

and

$$Cov(y) = Cov(X\beta + \epsilon) = \sigma^2 I_n = \underset{\text{Cov. matrix}}{\Sigma_{n \times n}}$$

→ Note that we have not made any distributional Assumption on ϵ yet.

→ we will introduce that bite later.

Q what is the expected value of cy? If c is Constant

Result :- we know.

$$E(y) = X\beta$$

$$E[cy] = cX\beta.$$

Now consider the Only Least Square estimator (OLS) estimator of β?

$$\boxed{\hat{\beta} = \underbrace{(X^T X)^{-1} X^T}_{\text{deterministic part}} y}$$

$$E(\hat{\beta}) = E((X^T X)^{-1} X^T y)$$

$$\Rightarrow \underbrace{(X^T X)^{-1} X^T}_{\text{Const}} E(y) = \underbrace{(X^T X)^{-1} X^T X}_{\text{Identity matrix}} \beta$$

$$\Rightarrow \beta$$

Result 2 :- OLS estimator $\hat{\beta}$ is an unbiased of β.

→ Supposed we are interested in some linear combination of regression coefficient, like $f(\beta) = C^T\beta$.

Result 3 :- Then the unbiased estimator of $C^T\beta$ is $C^T\hat{\beta}$, i.e.:

$$\boxed{E(C^T\hat{\beta}) = C^T\beta}$$

→ Suppose $C = X_0$ is a test point. Then we are interested in prediction $f(X_0) = X_0^T\beta$ are of this form.

$X_0^T\hat{\beta}$ is an unbiased estimator for the test point $C = X_0$.

Example :-
Suppose there are four Treatment :- Treat1, Treat2, Treat 3, and Treat4.

→ So the model consider is

$$y = \beta_0 + \beta_1 \text{Treat1} + \beta_2 \text{Treat2} + \beta_3 \text{Treat3} + e.$$

$E(y|Trt1) = \beta_0 + \beta_1$

$E(y|Trt2) = \beta_0 + \beta_2$

$E(y|Trt3) = \beta_0 + \beta_3$

$E(y|Trt4) = \beta_0$

→ Our regression coefficient are:

$$\beta = (\beta_0, \beta_1, \beta_2, \beta_3)^T.$$

→ We are interested in finding the diff b/w Treat 1 & 3

i.e.: $\boxed{\delta = \beta_1 - \beta_3}$

$E(y|Trt\,1) - E(y|Trt\,3) = \cancel{\beta_0} + \beta_1 - \cancel{\beta_0} + \beta_3$

$\Rightarrow \boxed{\beta_1 - \beta_3}$. good enough.

then we choose c as.

$$\boxed{c = (0, 1, 0, -1)}$$

i.e; $\delta = \beta_1 - \beta_3 = c^T \beta$.

So Gauss Markov Theorem tell us that $\hat{\delta} = c^T \hat{\beta}$ is the BLUE to estimate the effective diff b/w Trteat 1 and Treat 3.

Best linear unbiased estimator.

So $c^T \hat{\beta}$ is BLUE
for $c^T \beta = \beta_1 - \beta_3$

→ If we have any other linear Estimator $\bar{Q} = a^T y$ is unbiased for $c^T \beta$, that is

$$\boxed{E(a^T y) = c^T \beta}$$

then $\boxed{var(c^T \hat{\beta}) \leq var(a^T y)}$

OLS estimates of the parameter β have the smallest variance among all linear unbiased Estimates.

→ Consider the mean squared error (MSE) of an estimator $\tilde{\theta}$ in estimating θ :- target parameter
estimator

$$MSE(\tilde{\theta}) = E(\tilde{\theta} - \theta)^2$$
$$\Rightarrow Var(\tilde{\theta}) + E[(\tilde{\theta}) - \theta]^2$$
$$\Rightarrow Var(\tilde{\theta}) + \underbrace{[bias]^2}_{\to 0}$$

The Gauss-Markov Theorem implies that the least squares estimator has the smallest MSE among all the linear estimator with no bias

In short Explaination

Actual value

Population (time money) → $\boxed{Y_i = \alpha + \beta_1 x + \mu_i}$

α, β_1 : parameter.

↓ sampling method

(sample) → $\boxed{\hat{Y}_i = \hat{\alpha} + \hat{\beta} x + \mu_i}$

Estimated values.

α, β_1 → parameters.

OLS estimator : through. find value of α & β

then its BLUE — Estimator
B ↓ Best, L ↓ Linear, U ↓ unbiased

If we satisfied OLS Assumption than OLS estimator should be very efficient BLUE.

Linearity → $\bar{\alpha} + \bar{\beta}$. (no powers)

- Unbiasedness

population $\longleftrightarrow$ sample.
mean (μ) mean $\bar{X}$

$\boxed{E(\bar{x}) = \mu}$ → unbiased

$(E(\bar{x}) \neq \mu)$ → Biased
+ve -ve.

$\boxed{Y_i = \alpha + \beta X_i + \mu_i}$ — (1) First we proof unbiased.

By doing summation

$\Sigma Y_i = n\alpha + \beta_1 \Sigma X_i + \Sigma \mu_i$

Divide by n

$\frac{\Sigma Y_i}{n} = \frac{\cancel{n}\alpha}{\cancel{n}} + \beta_1 \frac{\Sigma X_i}{n} + \frac{\Sigma \mu_i}{n} \rightarrow 0.$

from Assumptn $\Sigma \mu = 0$.

$\boxed{\bar{Y} = \alpha + \beta_1 \bar{X}}$ — (2)

Substracting eqn (1) from (2).

$(Y_i - \bar{Y}) = (\alpha - \alpha) + \beta_1 (X_i - \bar{X}) + \mu_i$

Mean Deviatn

$\boxed{y_i = \beta_1 x_i + \mu_i}$

$Y_i = \alpha + \beta_1 X_i + \mu_i$

$\beta_1 = \frac{\Sigma x_i y_i}{\Sigma x_i^2}$

$\hat{\beta_1} = \frac{\Sigma x_i(\beta_1 x_i + \mu_i)}{\Sigma x_i^2} \Rightarrow \frac{\beta_1 \Sigma x_i^2 + \Sigma u_i x_i}{\Sigma x_i^2}$

$$\hat{\beta}_1 = \frac{\beta_1 \sum \cancel{x_i^2}}{\sum \cancel{x_i^2}} + \frac{\sum u_i x_i}{\sum x_i^2}.$$

$u_1 \rightarrow$ error term summation $= 0$.

$$\hat{\beta}_1 = \beta_1 + 0$$

$$E(\hat{\beta}_1) = \beta_1$$

$\downarrow$ $\downarrow$

sample = populatⁿ.

$\therefore$ unbiased.

[BLUE]

$\downarrow$

MVUE

(Min variance unbiased Estimator

Analysis of Variance (ANOVA)

Def^n :- ANOVA is a statistical Analysis or method used to compare the means of 2 or more group.

[Means of two group comparison] → [z test, t-test] used this specific test.

Categorical data → [chi-square test] → population propostion

ANOVA (parameters) Imp.

① Factors (variables)

② Levels (

eg :- Factor = Medicines.

Q- why we called medicines as factor?

∴ Because it has multi levels.

Medicines → factor.

5mg, 10mg, 20mg [Dosages] (levels).

eg $\boxed{\text{Mode of payment}}$ → factor

G Pay, Phone Pay, Paytm, Imps, Neft

Levels

Assumption in Anova

① **Normality of sample Distributⁿ Mean.**

* The distributⁿ of sample mean is normaly distributed.

② **Absence of outlier** ✓ imp.

* Outlying / outliers needs to be removed from dataset.

③ **Homogenity of variance** ✓ imp.

* each one of the population has same variance.

  eg $[\sigma_1^2 = \sigma_2^2 = \sigma_3^3]$

  population variance in different levels of each independent variable are equal.

④ **samples are independent and Random variab**

Types of ANOVA

One way ANOVA

Two-way ANOVA

(One factor with atleast 2 levels, these levels are Independent)

eg Doctor want to test a new medication to decrease headache, They split the participants in 3 conditions (10 mg, 20 mg, 30 mg). Doctor ask the participant to rate the headache [1-10].

Medication → Factor.

10mg   20mg   30mg ← levels (atleast 2 are there)

user 1 → 5   7   2.

user 2 → 9   8   7

(2). Two-way ANOVA)

Repeated Measures ANOVA)

one factor with atleast 2 levels, levels are dependent

eg   Running → Factor

Day 1 → Day 2 → Day 3 → levels.

user 1 → 8   5   7

2 → 7   4   3.

(3) Factorial ANOVA

* Two or more factors each of which atleast 2 levels, levels can be Independent or dependent.

eg:- $\boxed{\text{Running}}$ → Factors.

Day 1   Day 2   Day 3

$\boxed{\text{Gender}}$ → Factor

	Day 1	Day 2	Day 3
Men	8	5	6
Women	6	5	4

↓ levels.

Anova Test occurs in F- Distribution.

eg:- Hypothesis Test in ANOVA

① Null hypo. $H_0 \rightarrow \mu_1 = \mu_2 = \mu_3 \cdots = \mu_K$.

Alternate hypo $H_1 \rightarrow \mu_1 \neq \mu_2 \neq \mu_3 \cdots \neq \mu_K$.

or

H_1 = Atleast one of the Mean is not equal.

② Test statistics / F- Test

$$F = \frac{\text{Variance b/w sample}}{\text{Variance within sample}}$$ Imp

Variatn b/w sampl

eg

variance within sample

X_1	X_2	X_3
1	6	5
2	7	6
4	3	3
5	2	2
3	1	4
$\Sigma X_1 = 15$	$\Sigma X_2 = 19$	$\Sigma X_3 = 20$
$\bar{X}_1 = 3$	$\bar{X}_2 = 19/5$	$\bar{X}_3 = 20/5 = 4$

sample → mean.

$H_0 = \bar{X}_1 = \bar{X}_2 = \bar{X}_3$

H_1 = Atleast one sample is not equal.

F-Distribution → Right skewed Distributn

Chi-square → " " "

One ...

one ...

Ind...

One way ANOVA

One factor with atleast 2 # levels, levels are Independent.

(1) Doctor want to test a new medication which reduces headache. They split the participant into 3 condition [15mg, 30mg, 45mg]. Later on the doctor ask the patient to rate the headache b/w [1–10]. Are there any difference b/w 3 conditions? alpha = 0.05 ?

15mg	30 mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

Apply chi-sq. Test first to compare variance

compare Mean of Two Grp. wrt populatn proportion, chi-sq. Test.

(1) Define Null and Alternate hypothesis

$H_0 = \mu_{15} = \mu_{30} = \mu_{45}$

H_1 = Atleast one μ is not equal.

(2) state significance value

$\alpha = 0.05$ $\Rightarrow$ Conf. Interval = $(1-0.05)$ $\Rightarrow (0.95)$

(3) Calculate Degree of freedom

N = Total no. of sample

N = 21 . (7×3)

a = how many no. of categories are there w.r.t populatⁿ proportion?

a = 3 . (15 mg, 30 mg, 45 mg)

n = 7 (Records in each category)

degree of freedom

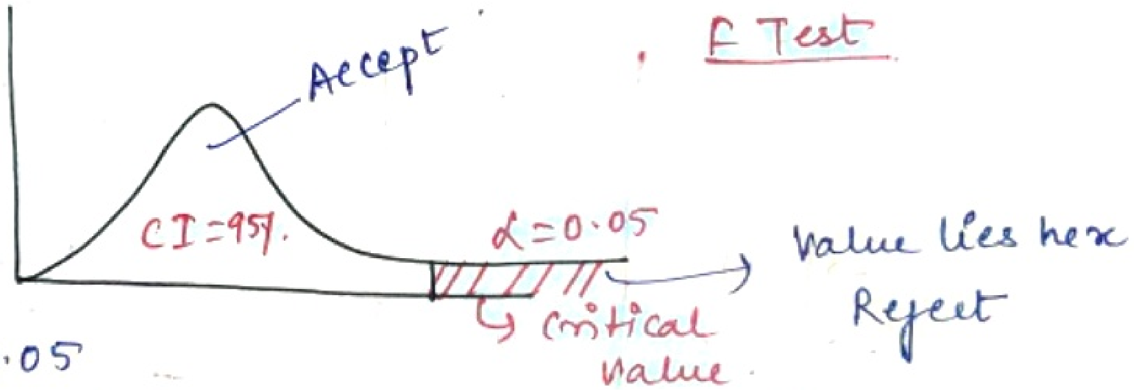
$\boxed{df_{between} = a-1 = 3-1 = 2}$

$\boxed{df_{within} = N-a = 21-3 = 18}$

$df_{total} = N-1 = 20$

(2, 18) ⇒ F Table
↓
significance or critical value

F Test



α = 0.05

(df_1, df_2)
(2, 18)

search on google F-Table

$\boxed{\text{Value} = 3.5546}$

Critical value = 3.5546

Decision Rule

If F is greater than 3.5546, we reject H_0 or fail to ~~Acce~~ Null hypo.

⑤ Compute F-Test statistics

$$F = \frac{\text{Variance b/w sample}}{\text{Variance within sample}}$$

Sum of sq. ↓ SS | degree of freed ↓ df | ↓ Mean square MS | F — f-Test

Between
Within
Total

15mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

$$SS \text{ between} = \Sigma\frac{(\Sigma a_i)^2}{n} - \frac{T^2}{N}.$$

(a_i) 15mg = 9 + 8 + 7 + 8 + 8 + 9 + 8 = 52

30mg = 47

45mg = 21

$$\Rightarrow \frac{(57)^2+(47)^2+(21)^2}{7} - \frac{T^2}{N} = \boxed{98.67}$$

Submission of all these three value

$$\Rightarrow \frac{(57)^2+(47)^2+(21)^2}{7} - \left[\frac{57+47+21}{21}\right]$$

	SS	df	(Mean Square) MS	F
Between	98.67	2	SS/df = 49.34	
Within	10.29	18	SS/df ⇒ 0.54	
Total	108.95	20		

~~86.56~~ 91.34

(2) SS within ⇒ $\sum y^2 - \frac{\sum(\sum a_i)^2}{n}$

$$\Rightarrow \sum y^2 = \left[\frac{57^2 + 47^2 + 21^2}{7}\right]$$

$$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 + 9^2 + 8^2 + 7^2 + 6^2 + 6^2 + 7^2 + 8^2 + 7^2 + 6^2 + 4^2 + 3^2 + 2^2 + 3^2 + 4^2 + 3^2 + 2^2$$

(sum of all sample points)

$\Rightarrow 853$

$$\Rightarrow 853 - \left[\frac{57^2 + 47^2 + 21^2}{7}\right]$$

$$\Rightarrow \boxed{10.29}$$

$SS_{Total} = 98.67 + 10.29 = 108.95$

$$F = \frac{\text{Variance b/w sample}}{\text{Variance within}}$$

Variance is nothing but mean squared

$$F = \frac{MS_{Between}}{MS_{within}} = \frac{49.34}{0.54} = \cancel{86.56}\ 91.34.$$

Conclusion here from F Test $86.56 > 3.55$, Reject Ho. Yes diff is there.

We are 95% confident about it.

Ordinary Least Square

The **least square method** is the process of finding the best-fitting curve or line of best fit for a set of data points by reducing the sum of the squares of the offsets (residual part) of the points from the curve. During the process of finding the relation between two variables, the trend of outcomes are estimated quantitatively. This process is termed as **regression analysis**. The method of curve fitting is an approach to regression analysis. This method of fitting equations which approximates the curves to given raw data is the least squares.

It is quite obvious that the fitting of curves for a particular data set are not always unique. Thus, it is required to find a curve having a minimal deviation from all the measured data points. This is known as the best-fitting curve and is found by using the least-squares method.

Least Square Method Definition

The least-squares method is a crucial statistical method that is practised to find a regression line or a best-fit line for the given pattern. This method is described by an equation with specific parameters. The method of least squares is generously used in evaluation and regression. In regression analysis, this method is said to be a standard approach for the approximation of sets of equations having more equations than the number of unknowns.

The method of least squares actually defines the solution for the minimization of the sum of squares of deviations or the errors in the result of each equation. Find the formula for sum of squares of errors, which help to find the variation in observed data.

The least-squares method is often applied in data fitting. The best fit result is assumed to reduce the sum of squared errors or residuals which are stated to be the differences between the observed or experimental value and corresponding fitted value given in the model.

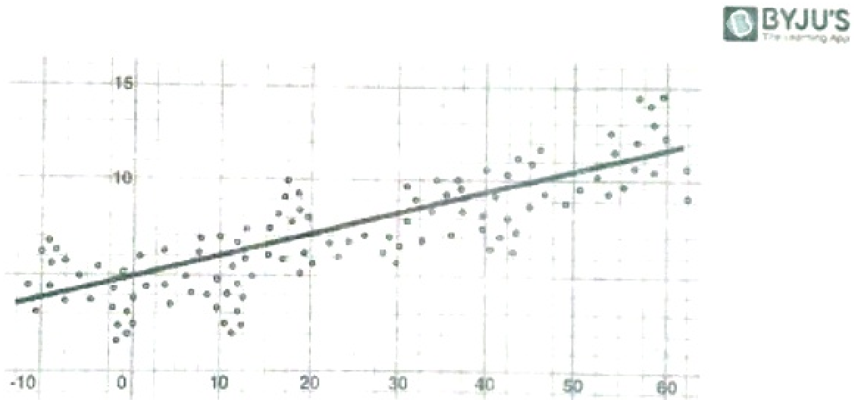
There are two basic categories of least-squares problems:

- Ordinary or linear least squares
- Nonlinear least squares

These depend upon linearity or nonlinearity of the residuals. The linear problems are often seen in regression analysis in statistics. On the other hand, the non-linear problems are generally used in the iterative method of refinement in which the model is approximated to the linear one with each iteration.

Least Square Method Graph

In linear regression, the line of best fit is a straight line as shown in the following diagram:



The given data points are to be minimized by the method of reducing residuals or offsets of each point from the line. The vertical offsets are generally used in surface, polynomial and hyperplane problems, while perpendicular offsets are utilized in common practice.

Vertical offsets Perpendicular offsets

Least Square Method Formula

The least-square method states that the curve that best fits a given set of observations, is said to be a curve having a minimum sum of the squared residuals (or deviations or errors) from the given data points. Let us assume that the given points of data are (x_1, y_1), (x_2, y_2), (x_3, y_3), …, (x_n, y_n) in which all x's are independent variables, while all y's are dependent ones. Also, suppose that f(x) is the fitting curve and d represents error or deviation from each given point.

Now, we can write:

$d_1 = y_1 - f(x_1)$

$d_2 = y_2 - f(x_2)$

$d_3 = y_3 - f(x_3)$

…..

$d_n = y_n - f(x_n)$

The least-squares explain that the curve that best fits is represented by the property that the sum of squares of all the deviations from given values must be minimum, i.e:

$Y = mx + c$

$a \quad b$

a

$\Sigma y = nm + c\Sigma x$

$$S = \sum_{i=1}^{n} d_i^{\,2}$$

$$S = \sum_{i=1}^{n} [y_i - f_{x_i}]^2$$

$$S = d_1^{\,2} + d_2^{\,2} + d_3^{\,2} + \cdots + d_n^{\,2}$$

Sum - Minimum Quantity

Suppose when we have to determine the equation of line of best fit for the given data, then we first use the following formula.

The equation of least square line is given by $Y = a + bX$

Normal equation for 'a':

$\sum Y = na + b\sum X$

Normal equation for 'b':

$\sum XY = a\sum X + b\sum X^2$

Solving these two normal equations we can get the required trend line equation.

Thus, we can get the line of best fit with formula $y = ax + b$

Solved Example

The Least Squares Model for a set of data (x_1, y_1), (x_2, y_2), (x_3, y_3), …, (x_n, y_n) passes through the point (x_a, y_a) where x_a is the average of the x_i's and y_a is the average of the y_i's. The below example explains how to find the equation of a straight line or a least square line using the least square method.

Question:

Consider the time series data given below:

x_i	8	3	2	10	11	3	6	5	6	8
y_i	4	12	1	12	9	4	9	6	1	14

Use the least square method to determine the equation of line of best fit for the data. Then plot the line.

Solution:

Mean of x_i values = (8 + 3 + 2 + 10 + 11 + 3 + 6 + 5 + 6 + 8)/10 = 62/10 = 6.2

Mean of y_i values = (4 + 12 + 1 + 12 + 9 + 4 + 9 + 6 + 1 + 14)/10 = 72/10 = 7.2

Straight line equation is y = a + bx.

The normal equations are

$\sum y = an + b\sum x$

$\sum xy = a\sum x + b\sum x^2$

x	y	x^2	xy
8	4	64	32
3	12	9	36
2	1	4	2
10	12	100	120

11	9	121	99
3	4	9	12
6	9	36	54
5	6	25	30
6	1	36	6
8	14	64	112
$\sum x = 62$	$\sum y = 72$	$\sum x^2 = 468$	$\sum xy = 503$

Substituting these values in the normal equations,

$10a + 62b = 72 \ldots (1)$

$62a + 468b = 503 \ldots (2)$

$(1) \times 62 - (2) \times 10,$

$620a + 3844b - (620a + 4680b) = 4464 - 5030$

$-836b = -566$

$b = 566/836$

$b = 283/418$

$b = 0.677$

Substituting $b = 0.677$ in equation (1),

$10a + 62(0.677) = 72$

10a + 41.974 = 72

10a = 72 - 41.974

10a = 30.026

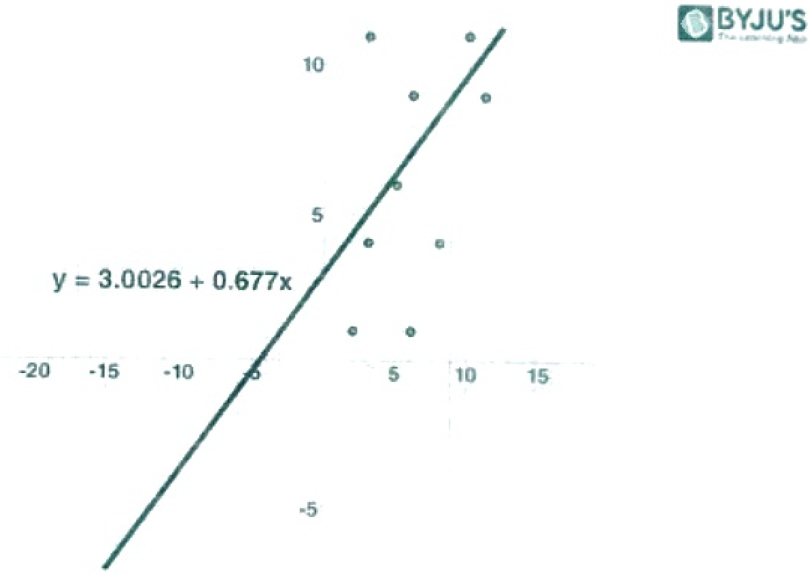
a = 30.026/10

a = 3.0026

Therefore, the equation becomes,

y = a + bx

y = 3.0026 + 0.677x



This is the required trend line equation.

Now, we can find the sum of squares of deviations from the obtained values as:

d_1 = [4 - (3.0026 + 0.677*8)] = (-4.4186)

d_2 = [12 - (3.0026 + 0.677*3)] = (6.9664)

d_3 = [1 - (3.0026 + 0.677*2)] = (-3.3566)

$d_4 = [12 - (3.0026 + 0.677*10)] = (2.2274)$

$d_5 = [9 - (3.0026 + 0.677*11)] = (-1.4496)$

$d_6 = [4 - (3.0026 + 0.677*3)] = (-1.0336)$

$d_7 = [9 - (3.0026 + 0.677*6)] = (1.9354)$

$d_8 = [6 - (3.0026 + 0.677*5)] = (-0.3876)$

$d_9 = [1 - (3.0026 + 0.677*6)] = (-6.0646)$

$d_{10} = [14 - (3.0026 + 0.677*8)] = (5.5814)$

$\sum d^2 = (-4.4186)^2 + (6.9664)^2 + (-3.3566)^2 + (2.2274)^2 + (-1.4496)^2 + (-1.0336)^2 + (1.9354)^2 + (-0.3876)^2 + (-6.0646)^2 + (5.5814)^2 = 159.27990$

Limitations for Least-Square Method

The least-squares method is a very beneficial method of curve fitting. Despite many benefits, it has a few shortcomings too. One of the main limitations is discussed here.

In the process of regression analysis, which utilizes the least-square method for curve fitting, it is inevitably assumed that the errors in the independent variable are negligible or zero. In such cases, when independent variable errors are non-negligible, the models are subjected to measurement errors. Therefore, here, the least square method may even lead to hypothesis testing, where parameter estimates and confidence intervals are taken into consideration due to the presence of errors occurring in the independent variables.

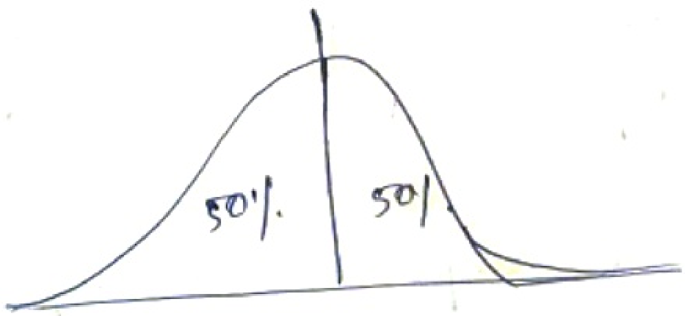
BOX - COX - TRANSFORMATIONS

Before we proceed to Box-Cox transformations. First we will know some terminologies.

① Power Law Distribution, Pareto Distribution.

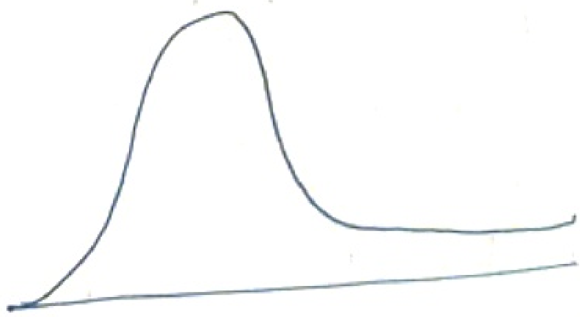
The power law (also called the scaling law), states that a relative change in one quantity results in a proportional change in another. The simplest example of law in action is square, if you double the length of a side (say from 2 to 4 inches) than the area will quadruple (from 4 to 16 inches sq).

①

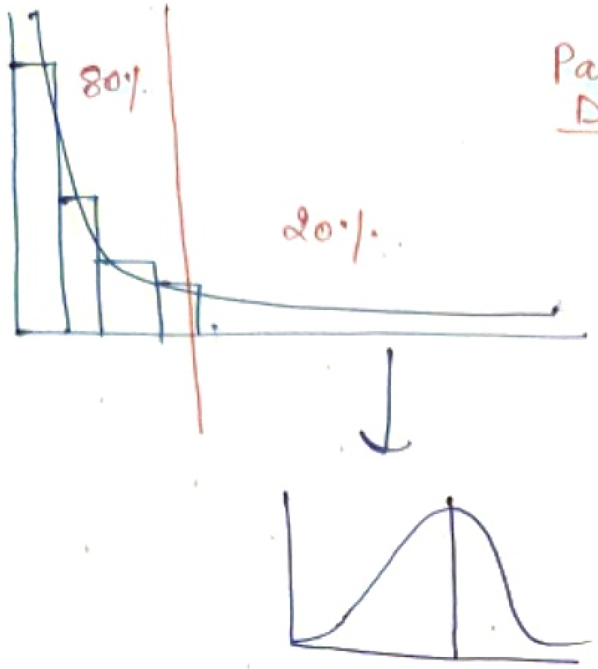


Normal Distribution

②



Right skewed
log normal Distri
$\ln(x) \rightarrow$ using we can make it Normal Distribution.



Pareto Distribution

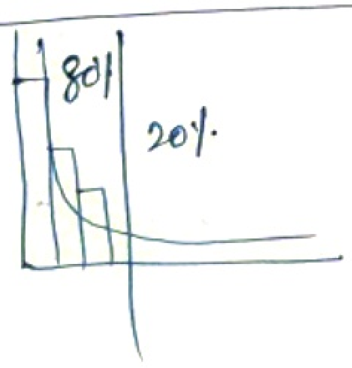
To Convert Pareto Distribution to Normal Distributⁿ we use a technique called

BOX- COX TRANSFORMA

Box - cox Transform

A Box-Cox transformation is a transformatⁿ of non-normal dependent variable into a normal shape

Normality is an ~~as~~ important assumption for many statiscally techniques.', if your data is not normal, applying a Box- cox means than you are able to run a broader no. of test.



At the core of the Box-COX transform is an exponent, lambda (λ), which varies from -5 to 5. All values of λ are Considered and the optimal value for your data is selected. "The optimal value" is the one which results in the best approximatⁿ of of a Normal Distributⁿ curve. The transform of Y has form.

$$y(\lambda) = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log y, & \text{if } \lambda = 0. \end{cases}$$

This test only work for positive data. However BOX-COX did propose a second formula that can be used for -ve y-values.

$$y(\lambda) = \begin{cases} \frac{(y+\lambda_2)^{\lambda_1}-1}{\lambda_1}, & \text{if } \lambda_1 \neq 0; \\ \log(y+\lambda_2), & \text{if } \lambda_1 = 0; \end{cases}$$

eg of pareto Distribution

1. 20% of Criminals Commit 80% of crimes.
2. 20% of drivers cause 80% of all traffic Accidents.
3. 80% of pollutⁿ originates from 20% of all factories
4. 20% of a Companies products represents 80% of sales.